

Examiner reports

Q1.

Again this question was done well by many students although some made minor errors. In part (c) a few students found the acceleration (as $1.4\mathbf{i} + 1.6\mathbf{j}$) and did not calculate the magnitude. Also, in part (d) some students found the angle as 41° rather than 49° . However many students gained full marks on this question.

Q2.

There were some very varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).

Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement method led candidates to the correct answer.

Q3.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like

$500 = \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j})^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q4.

This question was also done well by many candidates. In part (a), there were some confused responses and occasional problems with minus signs. Part (b) was also generally done well. In part (c), a few candidates had problems. The most common of these was to omit an arrow from the diagram to show the force; the other was to use the

lengths incorrectly when calculating the angle, for example $\tan^{-1}\left(\frac{8}{2}\right)$ and $\sin^{-1}\left(\frac{8}{8.25}\right)$.

Q5.

Parts (a), (b) and (c) of this question were often done very well. In part (a) candidates occasionally gave the answers "5" or "5j", but the vast majority of the answers were

correct. In part (b), many correctly stated the velocity, but some candidates then performed incorrect simplifications. Part (c) was also done well by those candidates who had obtained the velocity vector.

Part (d) proved to be more challenging. Some candidates realised that they needed to find the position vector and hence the bearing. These candidates usually made good progress, although some stopped when they had the position vector and others gave the answer as 52.3° . Some candidates tried to work with velocities.

Q6.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

Q7.

This was a popular starter question often yielding full marks, with only occasional careless errors in arithmetic in part (a) and signs in part (b).

Q8.

Parts (a) and (b) were done well by the better prepared candidates, but few spotted and used the pattern of decreasing distances in part (c).

Q9.

Part (a) was done very well by a good number of the candidates. A few worked with the magnitudes of the quantities, starting with the magnitude of the force, rather than with the actual vectors. Part (b) was found to be much more demanding. The two main errors made by the candidates were to assume that the particle still had the same acceleration as in part (a) or to use a time of 45 s rather than 25 s for the second stage of the motion.

Q10.

A very small number of candidates seemed unsure how to deal with the vectors and made very little progress with the question. The majority of candidates did produce quite good solutions. Part (a) did not cause any particular difficulties. In part (b), some

candidates wrote $\sqrt{9^2 - 4^2}$ instead of $\sqrt{9^2 + 4^2}$. Those who could visualise part (c) had no difficulty, but some found it hard to identify a triangle to work in and tried to use the sine or cosine rules. A few candidates produced incorrect expressions involving sin or cos instead of using tan.